Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: J & J De Raadt Dairy

Physical address of dairy:

6460 Smith RD	Oakdale	Stanislaus	95361
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X010-X017-X069-XXXX

B. OPERATORS

DeRaadt, John and or Joanne

Operator name: DeRaadt, John and or Joanne

Telephone no.: (209) 845-1572 (Landline) (Cellular)

6460 Smith RD	Oakdale	CA	95361
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

DeRaadt, John and or Joanne

Legal owner name: DeRaadt, John and or Joanne

Telephone no.: (209) 845-1572 (Landline) (Cellular)

6460 Smith RD	Oakdale	CA	95361
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	140	200	400	250	0
Number under roof	950	0	0	0	0	0
Maximum number	950	140	200	400	250	0
Average number	950	140	200	400	250	0
Avg live weight (lbs)	1,375	1,425	900	700		

Predominant milk cow breed: Other

Average milk production: 73 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 33,415.98 *tons per reporting period*

Total nitrogen from manure: 418,249.31 *lbs per reporting period*

After ammonia losses (30% loss applied): 292,774.52 *lbs per reporting period*

Total phosphorus from manure: 68,655.03 *lbs per reporting period*

Total potassium from manure: 183,806.22 *lbs per reporting period*

Total salt from manure: 479,500.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 8,745,000 *gallons*

Total nitrogen generated: 38,508.09 *lbs*

Total phosphorus generated: 5,305.58 *lbs*

Total potassium generated: 82,167.12 *lbs*

Total salt generated: 497,545.59 *lbs*

	8,745,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	8,745,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal OID	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/03/2023	Solid commercial fertilizer *4-0-0*	2.30 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000
07/24/2023	Solid commercial fertilizer *UN 32*	1.15 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	919.08	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	919.08	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	Corral solids	714.00 *ton*	Dry-weight	55.6		18,600.00	6,830.00	24,900.00		56.90

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	11,793.00	4,330.44	15,787.40	360,764.21
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	11,793.00	4,330.44	15,787.40	360,764.21

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	40	40	2	process wastewater	X010-X017-X069-XXXX
Field 2	5	5	1	manure	X010-X017-X023-XXXX
Field 3	15	15	2	manure	X010-X017-X069-XXXX
Field 4	8	8	2	manure	X010-X017-X069-XXXX
Field 5	18	18	1	manure	X010-X017-X069-XXXX
Field 7	40	40	2	process wastewater	X010-X016-X015-XXXX
Field 8	7	7	2	process wastewater	X010-X017-X069-XXXX
Totals for areas that were used for application	133	133	12		
Totals for areas that were not used for application					
Land application area totals	133	133	12		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	674.18 *ton*	As-is		75.6	4,370.00	1,050.00	15,100.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.85	147.31	35.39	509.01	970.55

Field 1

07/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 07/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	1,024.00 *ton*	As-is		67.2	4,230.00	340.00	7,450.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.60	216.58	17.41	381.44	990.82

Field 2

Field name: Field 2

01/01/2023: Pasture

Crop: Pasture Acres planted: 5 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	75.00 *ton*	As-is		70.0	8,170.00	830.00	6,910.00		9.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	276.00	39.00	283.50	0.00
Total actual harvest content	15.00	245.10	24.90	207.30	815.40

Field 3

Field name: Field 3

Field 3

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	253.50 *ton*	As-is		74.4	4,040.00	1,000.00	11,700.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.90	136.55	33.80	395.46	917.20

07/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 07/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	384.00 *ton*	As-is		71.2	4,150.00	520.00	5,090.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.60	212.48	26.62	260.61	766.77

Field 4

Field name: Field 4

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	135.20 *ton*	As-is		74.4	4,040.00	1,000.00	11,700.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.90	136.55	33.80	395.46	917.20

Field 4

07/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 07/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	204.80 *ton*	As-is		71.2	4,150.00	520.00	5,090.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.60	212.48	26.62	260.61	766.77

Field 5

Field name: Field 5

01/01/2023: Pasture

Crop: Pasture Acres planted: 18 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	270.00 *ton*	As-is		70.0	8,170.00	830.00	6,910.00		9.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	276.00	39.00	283.50	0.00
Total actual harvest content	15.00	245.10	24.90	207.30	815.40

Field 7

Field name: Field 7

Field 7

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	674.18 *ton*	As-is		71.6	3,690.00	820.00	7,900.00		9.46

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.85	124.39	27.64	266.30	905.64

07/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 07/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	1,024.00 *ton*	As-is		71.8	4,060.00	560.00	5,160.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.60	207.87	28.67	264.19	750.80

Field 8

Field name: Field 8

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 7 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	118.30 *ton*	As-is		74.4	4,040.00	1,000.00	11,700.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.90	136.55	33.80	395.46	917.20

Field 8

07/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 07/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	179.20 *ton*	As-is		71.2	4,150.00	520.00	5,090.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.60	212.48	26.62	260.61	766.77

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/01/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	184.21	29.91	255.81	1,516.62	1,560,000.00 *gal*
Application event totals		184.21	29.91	255.81	1,516.62	

Field 1 - 07/03/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 07/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	121.72	18.81	300.17	1,476.06	1,320,000.00 *gal*
Canal OID	Ground water	0.70	0.00	0.00	35.80	3,300,000.00 *gal*
Application event totals		122.42	18.81	300.17	1,511.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.07	0.00	0.00	55.33	5,100,000.00 *gal*
Application event totals		1.07	0.00	0.00	55.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.07	0.00	0.00	55.33	5,100,000.00 *gal*
Application event totals		1.07	0.00	0.00	55.33	

Field 1 - 07/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	150.89	14.59	407.44	2,838.97	1,260,000.00 *gal*
Canal OID	Ground water	0.66	0.00	0.00	34.17	3,150,000.00 *gal*
Application event totals		151.55	14.59	407.44	2,873.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.04	0.00	0.00	53.70	4,950,000.00 *gal*
Application event totals		1.04	0.00	0.00	53.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.01	0.00	0.00	52.07	4,800,000.00 *gal*
Application event totals		1.01	0.00	0.00	52.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.95	0.00	0.00	48.82	4,500,000.00 *gal*
Application event totals		0.95	0.00	0.00	48.82	

Field 2 - 01/01/2023: Pasture

Field name: Field 2

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.95	0.00	0.00	48.82	562,500.00 *gal*
Application event totals		0.95	0.00	0.00	48.82	

Field 2 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.88	0.00	0.00	45.56	525,000.00 *gal*
Application event totals		0.88	0.00	0.00	45.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	264.27	97.04	353.78	8,084.35	80.00 *ton*
Application event totals		264.27	97.04	353.78	8,084.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.01	0.00	0.00	52.07	600,000.00 *gal*
Application event totals		1.01	0.00	0.00	52.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.88	0.00	0.00	45.56	525,000.00 *gal*
Application event totals		0.88	0.00	0.00	45.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.01	0.00	0.00	52.07	600,000.00 *gal*
Application event totals		1.01	0.00	0.00	52.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.95	0.00	0.00	48.82	562,500.00 *gal*
Application event totals		0.95	0.00	0.00	48.82	

Field 2 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.88	0.00	0.00	45.56	525,000.00 *gal*
Application event totals		0.88	0.00	0.00	45.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.95	0.00	0.00	48.82	562,500.00 *gal*
Application event totals		0.95	0.00	0.00	48.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.95	0.00	0.00	48.82	562,500.00 *gal*
Application event totals		0.95	0.00	0.00	48.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.88	0.00	0.00	45.56	525,000.00 *gal*
Application event totals		0.88	0.00	0.00	45.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.88	0.00	0.00	45.56	525,000.00 *gal*
Application event totals		0.88	0.00	0.00	45.56	

Field 3 - 11/01/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 3 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	176.78	74.55	334.76	3,197.13	130.00 *ton*
Application event totals		176.78	74.55	334.76	3,197.13	

Field 3 - 07/03/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 07/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	242.25	88.95	324.30	7,410.66	220.00 *ton*
Application event totals		242.25	88.95	324.30	7,410.66	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.10	0.00	0.00	56.41	1,950,000.00 *gal*
Application event totals		1.10	0.00	0.00	56.41	

07/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.01	0.00	0.00	52.07	1,800,000.00 *gal*
Application event totals		1.01	0.00	0.00	52.07	

Field 3 - 07/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.01	0.00	0.00	52.07	1,800,000.00 *gal*
Application event totals		1.01	0.00	0.00	52.07	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.01	0.00	0.00	52.07	1,800,000.00 *gal*
Application event totals		1.01	0.00	0.00	52.07	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.97	0.00	0.00	49.90	1,725,000.00 *gal*
Application event totals		0.97	0.00	0.00	49.90	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.97	0.00	0.00	49.90	1,725,000.00 *gal*
Application event totals		0.97	0.00	0.00	49.90	

09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.93	0.00	0.00	47.73	1,650,000.00 *gal*
Application event totals		0.93	0.00	0.00	47.73	

Field 4 - 11/01/2022: Oats, silage-soft dough

Field 4 - 11/01/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	178.48	75.27	337.98	3,227.88	70.00 *ton*
Application event totals		178.48	75.27	337.98	3,227.88	

Field 4 - 07/04/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 07/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	227.11	83.39	304.03	6,947.49	110.00 *ton*
Application event totals		227.11	83.39	304.03	6,947.49	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.11	0.00	0.00	56.95	1,050,000.00 *gal*
Application event totals		1.11	0.00	0.00	56.95	

07/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Field 4 - 07/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.03	0.00	0.00	52.89	975,000.00 *gal*
Application event totals		1.03	0.00	0.00	52.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.03	0.00	0.00	52.89	975,000.00 *gal*
Application event totals		1.03	0.00	0.00	52.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.03	0.00	0.00	52.89	975,000.00 *gal*
Application event totals		1.03	0.00	0.00	52.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.03	0.00	0.00	52.89	975,000.00 *gal*
Application event totals		1.03	0.00	0.00	52.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.03	0.00	0.00	52.89	975,000.00 *gal*
Application event totals		1.03	0.00	0.00	52.89	

Field 4 - 07/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.95	0.00	0.00	48.82	900,000.00 *gal*
Application event totals		0.95	0.00	0.00	48.82	

Field 5 - 01/01/2023: Pasture

Field name: Field 5

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.98	0.00	0.00	50.63	2,100,000.00 *gal*
Application event totals		0.98	0.00	0.00	50.63	

05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.98	0.00	0.00	50.63	2,100,000.00 *gal*
Application event totals		0.98	0.00	0.00	50.63	

06/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	275.28	101.08	368.52	8,421.20	300.00 *ton*
Application event totals		275.28	101.08	368.52	8,421.20	

06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.95	0.00	0.00	48.82	2,025,000.00 *gal*
Application event totals		0.95	0.00	0.00	48.82	

Field 5 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.91	0.00	0.00	47.01	1,950,000.00 *gal*
Application event totals		0.91	0.00	0.00	47.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.95	0.00	0.00	48.82	2,025,000.00 *gal*
Application event totals		0.95	0.00	0.00	48.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.95	0.00	0.00	48.82	2,025,000.00 *gal*
Application event totals		0.95	0.00	0.00	48.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.91	0.00	0.00	47.01	1,950,000.00 *gal*
Application event totals		0.91	0.00	0.00	47.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.95	0.00	0.00	48.82	2,025,000.00 *gal*
Application event totals		0.95	0.00	0.00	48.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.91	0.00	0.00	47.01	1,950,000.00 *gal*
Application event totals		0.91	0.00	0.00	47.01	

Field 5 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.91	0.00	0.00	47.01	1,950,000.00 *gal*
Application event totals		0.91	0.00	0.00	47.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.91	0.00	0.00	47.01	1,950,000.00 *gal*
Application event totals		0.91	0.00	0.00	47.01	

Field 7 - 11/01/2022: Oats, silage-soft dough

Field name: Field 7

Crop: Oats, silage-soft dough Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	155.87	25.31	216.45	1,283.29	1,320,000.00 *gal*
Application event totals		155.87	25.31	216.45	1,283.29	

Field 7 - 07/04/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage Plant date: 07/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 7 - 07/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	121.72	18.81	300.17	1,476.06	1,320,000.00 *gal*
Canal OID	Ground water	0.66	0.00	0.00	34.17	3,150,000.00 *gal*
Application event totals		122.38	18.81	300.17	1,510.24	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.07	0.00	0.00	55.33	5,100,000.00 *gal*
Application event totals		1.07	0.00	0.00	55.33	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.04	0.00	0.00	53.70	4,950,000.00 *gal*
Application event totals		1.04	0.00	0.00	53.70	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.04	0.00	0.00	53.70	4,950,000.00 *gal*
Application event totals		1.04	0.00	0.00	53.70	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	150.89	14.59	407.44	2,838.97	1,260,000.00 *gal*
Canal OID	Ground water	0.66	0.00	0.00	34.17	3,150,000.00 *gal*
Application event totals		151.55	14.59	407.44	2,873.14	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.01	0.00	0.00	52.07	4,800,000.00 *gal*
Application event totals		1.01	0.00	0.00	52.07	

Field 7 - 07/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.98	0.00	0.00	50.45	4,650,000.00 *gal*
Application event totals		0.98	0.00	0.00	50.45	

Field 8 - 11/01/2022: Oats, silage-soft dough

Field name: Field 8

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	161.94	26.29	224.89	1,333.29	240,000.00 *gal*
Application event totals		161.94	26.29	224.89	1,333.29	

Field 8 - 07/04/2023: Corn, silage

Field name: Field 8

Crop: Corn, silage

Plant date: 07/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	126.46	19.54	311.86	1,533.57	240,000.00 *gal*
Canal OID	Ground water	0.72	0.00	0.00	37.19	600,000.00 *gal*
Application event totals		127.18	19.54	311.86	1,570.77	

Field 8 - 07/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.08	0.00	0.00	55.79	900,000.00 *gal*
Application event totals		1.08	0.00	0.00	55.79	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.08	0.00	0.00	55.79	900,000.00 *gal*
Application event totals		1.08	0.00	0.00	55.79	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	153.97	14.89	415.76	2,896.91	225,000.00 *gal*
Canal OID	Ground water	0.68	0.00	0.00	34.87	562,500.00 *gal*
Application event totals		154.64	14.89	415.76	2,931.78	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.08	0.00	0.00	55.79	900,000.00 *gal*
Application event totals		1.08	0.00	0.00	55.79	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.08	0.00	0.00	55.79	900,000.00 *gal*
Application event totals		1.08	0.00	0.00	55.79	

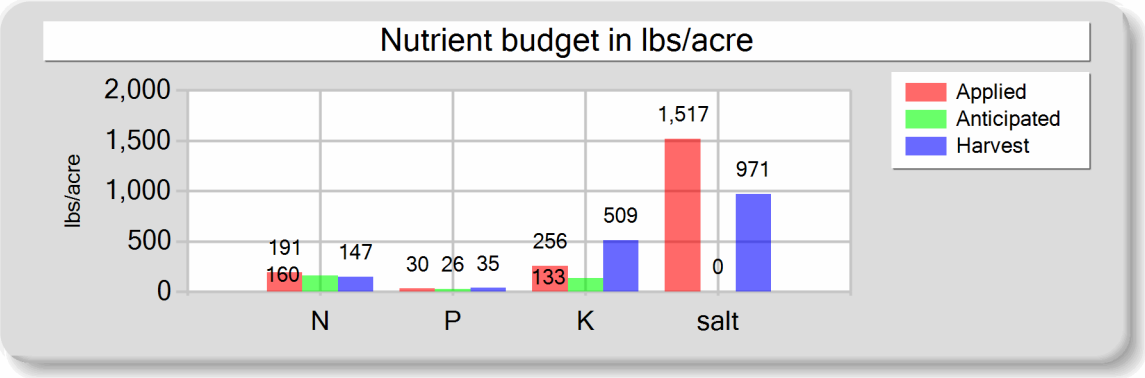
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.99	0.00	0.00	51.14	825,000.00 *gal*
Application event totals		0.99	0.00	0.00	51.14	

B. NUTRIENT BUDGET

Field 1 - 11/01/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	184.21	29.91	255.81	1,516.62
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.21	29.91	255.81	1,516.62
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	147.31	35.39	509.01	970.55
Nutrient balance	43.90	-5.49	-253.20	546.07
Applied to removed ratio	1.30	0.85	0.50	1.56

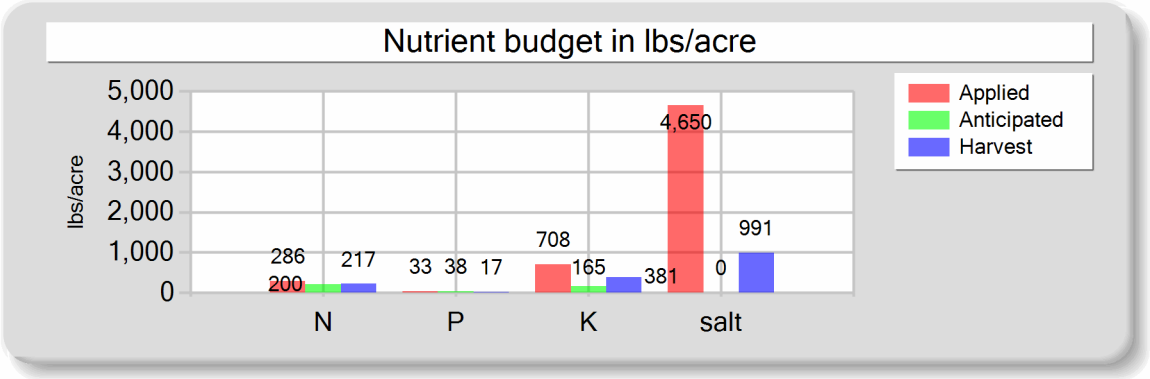
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,560,000.00	*gallons*
57.45	*acre-inches*
1.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 07/03/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 07/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	272.61	33.40	707.61	4,315.03
Fresh water	6.51	0.00	0.00	335.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.12	33.40	707.61	4,650.25
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	216.58	17.41	381.44	990.82
Nutrient balance	69.54	15.99	326.17	3,659.43
Applied to removed ratio	1.32	1.92	1.86	4.69

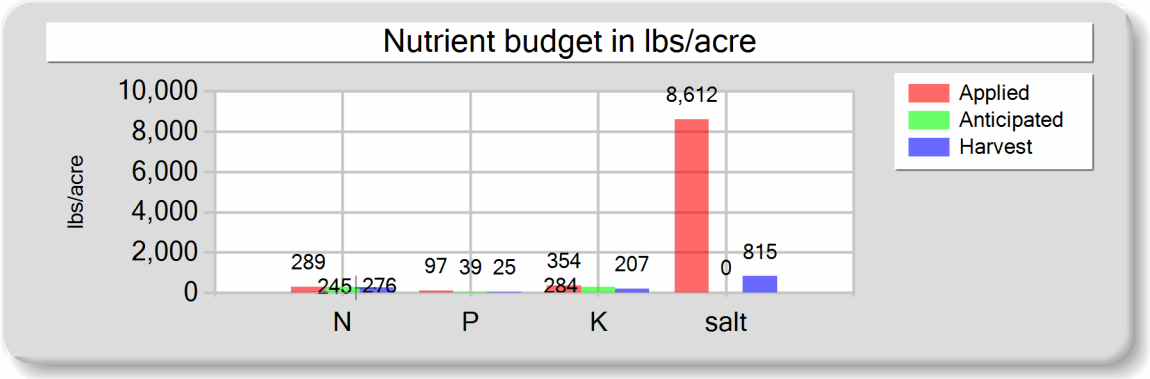
Fresh water applied	
30,900,000.00	*gallons*
1,137.94	*acre-inches*
28.45	*inches/acre*

Process wastewater applied	
2,580,000.00	*gallons*
95.01	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 01/01/2023: Pasture

Field name: Field 2 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	264.27	97.04	353.78	8,084.35
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10.24	0.00	0.00	527.24
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	288.51	97.04	353.78	8,611.59
Anticipated crop nutrient removal	276.00	39.00	283.50	0.00
Actual crop nutrient removal	245.10	24.90	207.30	815.40
Nutrient balance	43.41	72.14	146.48	7,796.19
Applied to removed ratio	1.18	3.90	1.71	10.56

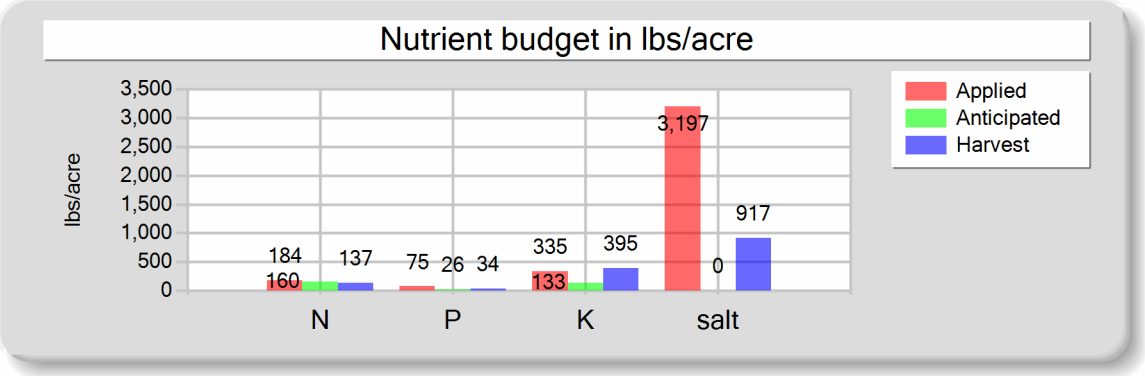
Fresh water applied
6,075,000.00 *gallons*
223.72 *acre-inches*
44.74 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 3 - 11/01/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	176.78	74.55	334.76	3,197.13
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.78	74.55	334.76	3,197.13
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	136.55	33.80	395.46	917.20
Nutrient balance	47.23	40.75	-60.70	2,279.94
Applied to removed ratio	1.35	2.21	0.85	3.49

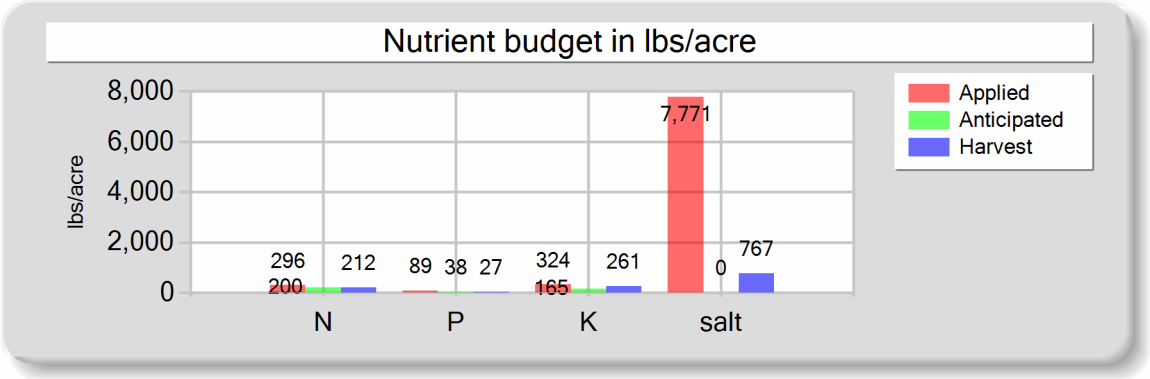
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 3 - 07/03/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 07/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	242.25	88.95	324.30	7,410.66
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.00	0.00	0.00	360.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	296.24	88.95	324.30	7,770.83
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	212.48	26.62	260.61	766.77
Nutrient balance	83.76	62.33	63.69	7,004.06
Applied to removed ratio	1.39	3.34	1.24	10.13

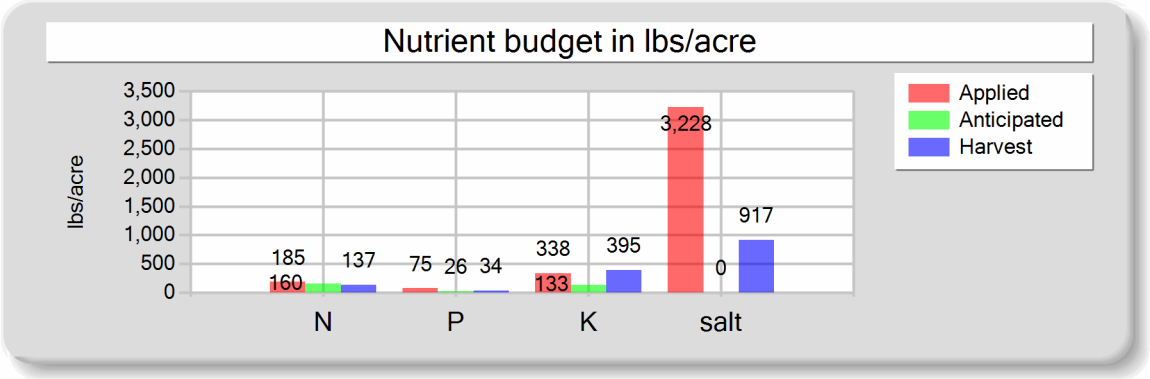
Fresh water applied	
12,450,000.00	*gallons*
458.49	*acre-inches*
30.57	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/01/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.48	75.27	337.98	3,227.88
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.48	75.27	337.98	3,227.88
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	136.55	33.80	395.46	917.20
Nutrient balance	48.93	41.47	-57.48	2,310.68
Applied to removed ratio	1.36	2.23	0.85	3.52

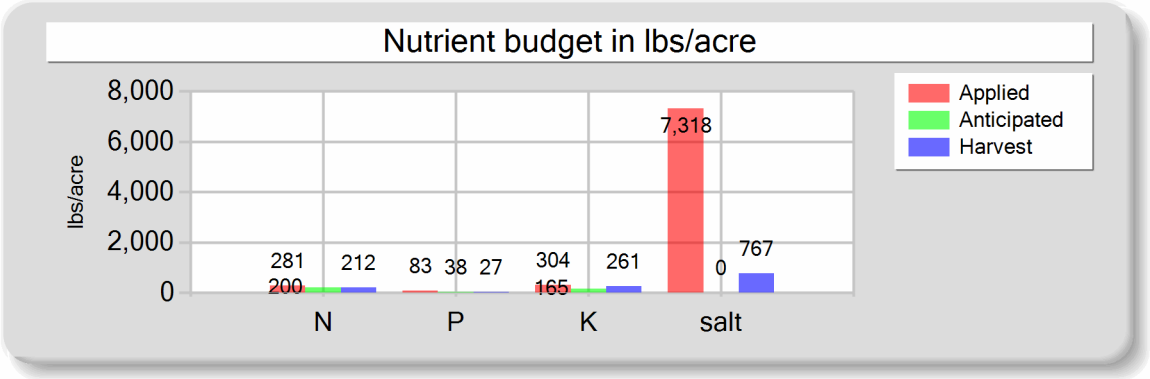
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 4 - 07/04/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 07/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	227.11	83.39	304.03	6,947.49
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.19	0.00	0.00	370.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	281.30	83.39	304.03	7,317.70
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	212.48	26.62	260.61	766.77
Nutrient balance	68.82	56.77	43.42	6,550.92
Applied to removed ratio	1.32	3.13	1.17	9.54

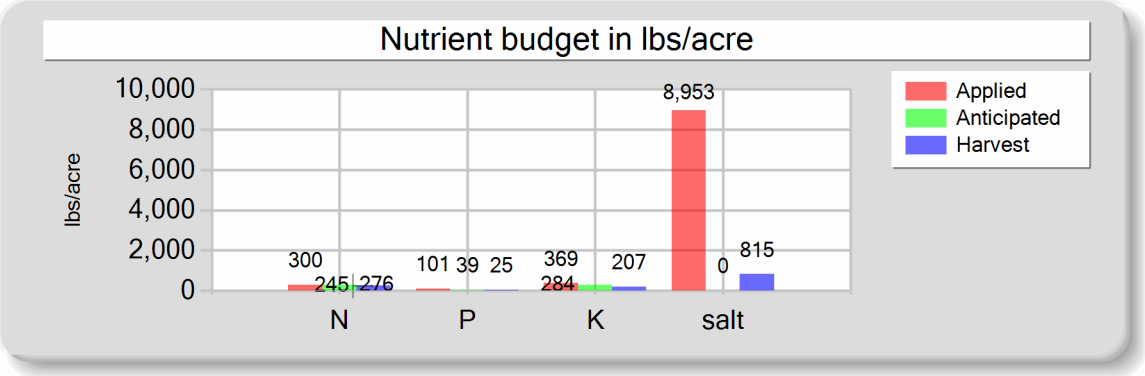
Fresh water applied	
6,825,000.00	*gallons*
251.34	*acre-inches*
31.42	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 01/01/2023: Pasture

Field name: Field 5 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	275.28	101.08	368.52	8,421.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10.32	0.00	0.00	531.58
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	299.60	101.08	368.52	8,952.78
Anticipated crop nutrient removal	276.00	39.00	283.50	0.00
Actual crop nutrient removal	245.10	24.90	207.30	815.40
Nutrient balance	54.50	76.18	161.22	8,137.38
Applied to removed ratio	1.22	4.06	1.78	10.98

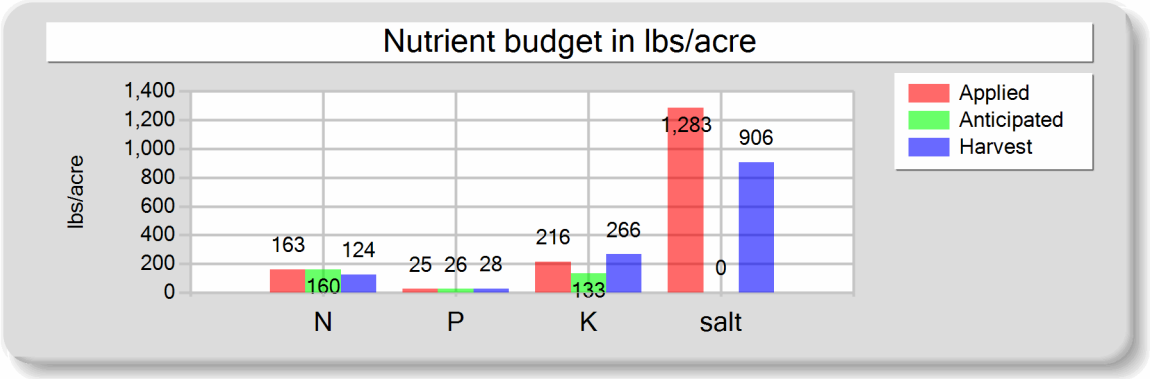
Fresh water applied	
22,050,000.00	*gallons*
812.03	*acre-inches*
45.11	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 11/01/2022: Oats, silage-soft dough

Field name: Field 7 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	155.87	25.31	216.45	1,283.29
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	162.87	25.31	216.45	1,283.29
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	124.39	27.64	266.30	905.64
Nutrient balance	38.48	-2.33	-49.85	377.65
Applied to removed ratio	1.31	0.92	0.81	1.42

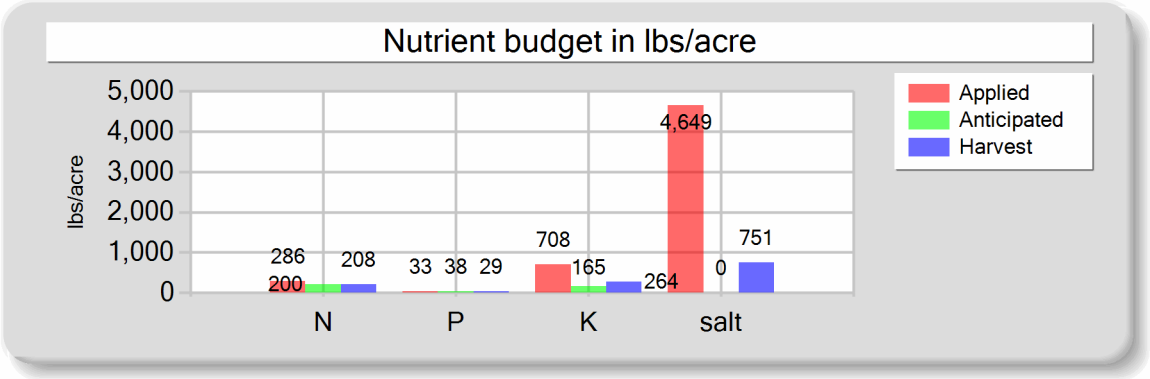
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,320,000.00	*gallons*
48.61	*acre-inches*
1.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 07/04/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 07/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	272.61	33.40	707.61	4,315.03
Fresh water	6.48	0.00	0.00	333.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.09	33.40	707.61	4,648.62
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	207.87	28.67	264.19	750.80
Nutrient balance	78.21	4.73	443.42	3,897.83
Applied to removed ratio	1.38	1.16	2.68	6.19

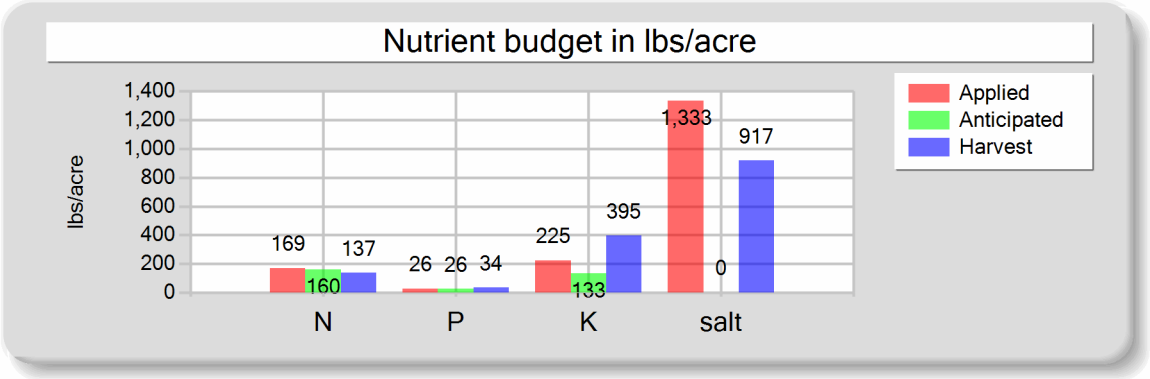
Fresh water applied	
30,750,000.00	*gallons*
1,132.42	*acre-inches*
28.31	*inches/acre*

Process wastewater applied	
2,580,000.00	*gallons*
95.01	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 11/01/2022: Oats, silage-soft dough

Field name: Field 8 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	161.94	26.29	224.89	1,333.29
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	168.94	26.29	224.89	1,333.29
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	136.55	33.80	395.46	917.20
Nutrient balance	32.39	-7.51	-170.57	416.10
Applied to removed ratio	1.24	0.78	0.57	1.45

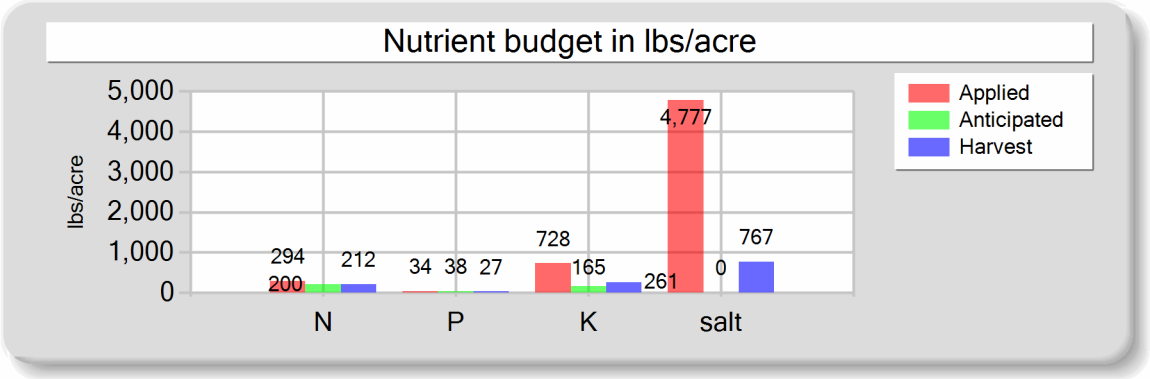
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
240,000.00 *gallons*
8.84 *acre-inches*
1.26 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 8 - 07/04/2023: Corn, silage

Field name: Field 8 Crop: Corn, silage Plant date: 07/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	280.43	34.43	727.62	4,430.48
Fresh water	6.73	0.00	0.00	346.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.16	34.43	727.62	4,776.86
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	212.48	26.62	260.61	766.77
Nutrient balance	81.68	7.80	467.02	4,010.09
Applied to removed ratio	1.38	1.29	2.79	6.23

Fresh water applied	
5,587,500.00	*gallons*
205.77	*acre-inches*
29.40	*inches/acre*

Process wastewater applied	
465,000.00	*gallons*
17.12	*acre-inches*
2.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,500.00	9,910.00	44,500.00							42.50
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	6,830.00	24,900.00							56.90
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 04/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 39.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,650.00	1,630.00	8,070.00							77.90
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 49.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	8,250.00	38,400.00							45.80
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/11/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 42.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,300.00	4,410.00	20,900.00							47.90
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

1QWW

Sample and source description: 1QWW

Sample date: 02/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	566.00	199.00	0.00	0.00	91.90	786.00								5,900.00	4,660
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	442.00	300.00	0.00	0.00	68.30	1,090.00								6,990.00	5,360
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	574.00	381.00	0.00	0.00	55.50	1,550.00								9,300.00	10,800
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 11/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,220.00	515.00	0.00	0.00	119.00	1,110.00								6,300.00	7,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal OID

Canal OID

Fresh Water

Sample description: Fresh Water

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.01	0.00	0.00								70.90	52
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

Field 1

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			123.00				
DL			4.00				

Field 2

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			119.00				
DL			4.00				

Field 3

Field 3

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			123.00				
DL			4.00				

Field 4

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			123.00				
DL			4.00				

Field 5

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			119.00				
DL			4.00				

Field 7

Field 7

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			119.00				
DL			4.00				

Field 8

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			123.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Field 1 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,370.00	1,050.00	15,100.00		11.80
DL	100.00	100.00	100.00		0.01

Field 1 - 07/03/2023: Corn, silage

Field 1 - 07/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,230.00	340.00	7,450.00		5.90
DL	100.00	100.00	100.00		0.01

Field 2 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,170.00	830.00	6,910.00		9.06
DL	100.00	100.00	100.00		0.01

Field 3 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,040.00	1,000.00	11,700.00		10.60
DL	100.00	100.00	100.00		0.01

Field 3 - 07/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,150.00	520.00	5,090.00		5.20
DL	100.00	100.00	100.00		0.01

Field 4 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,040.00	1,000.00	11,700.00		10.60
DL	100.00	100.00	100.00		0.01

Field 4 - 07/04/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,150.00	520.00	5,090.00		5.20
DL	100.00	100.00	100.00		0.01

Field 5 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,170.00	830.00	6,910.00		9.06
DL	100.00	100.00	100.00		0.01

Field 7 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,690.00	820.00	7,900.00		9.46
DL	100.00	100.00	100.00		0.01

Field 7 - 07/04/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,060.00	560.00	5,160.00		5.20
DL	100.00	100.00	100.00		0.01

Field 8 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,040.00	1,000.00	11,700.00		10.60
DL	100.00	100.00	100.00		0.01

Field 8 - 07/04/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,150.00	520.00	5,090.00		5.20
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

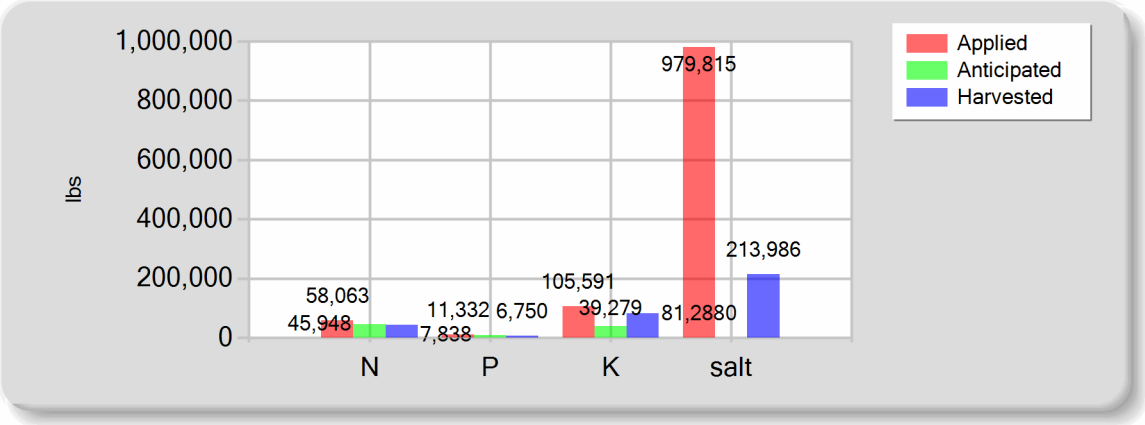
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

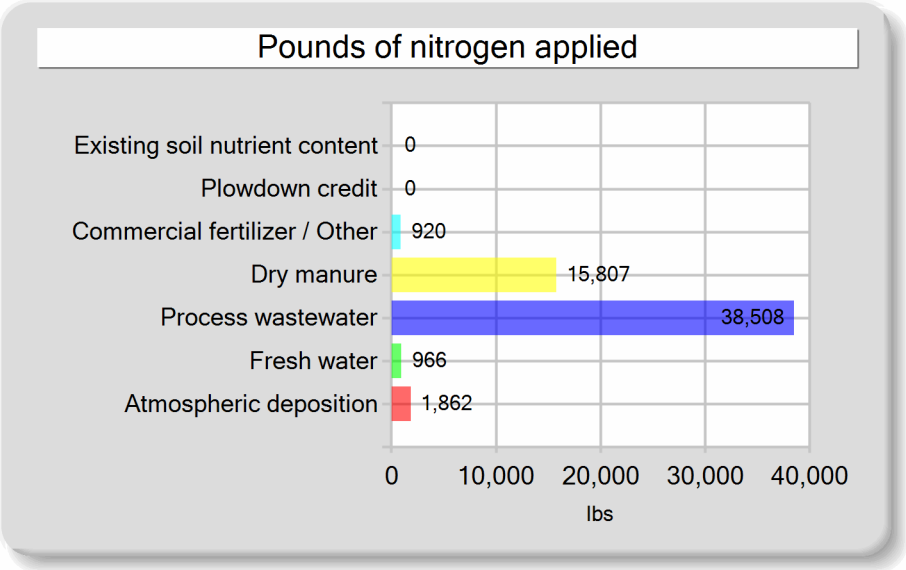
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

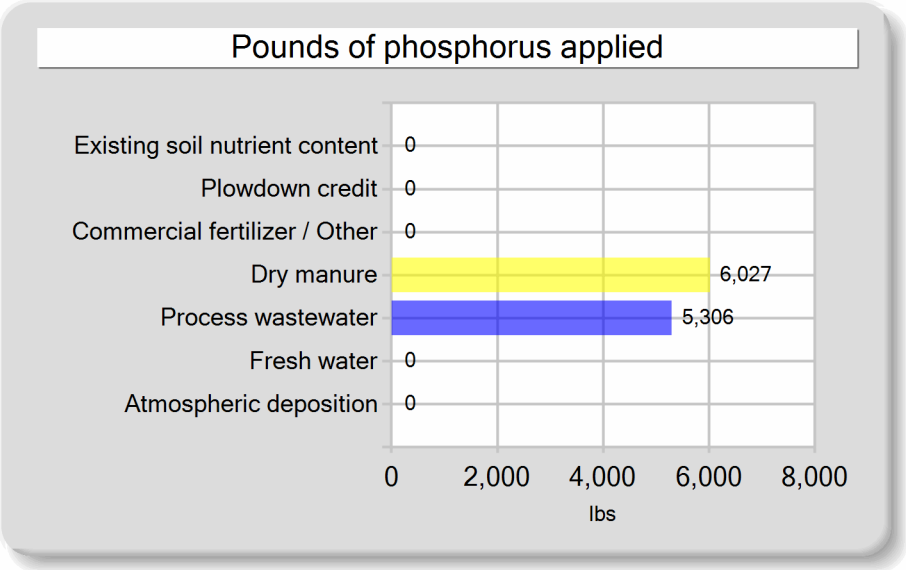
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	920.00	0.00	0.00	0.00
Dry manure	15,806.53	6,026.55	23,424.15	432,523.12
Process wastewater	38,508.09	5,305.58	82,167.12	497,545.59
Fresh water	966.22	0.00	0.00	49,745.80
Atmospheric deposition	1,862.00	0.00	0.00	0.00
Total nutrients applied	58,062.83	11,332.13	105,591.28	979,814.51
Anticipated crop nutrient removal	45,948.00	7,838.00	39,278.50	0.00
Actual crop nutrient removal	43,953.96	6,750.05	81,287.50	213,985.57
Nutrient balance	14,108.87	4,582.08	24,303.78	765,828.94
Applied to removed ratio	1.32	1.68	1.30	4.58

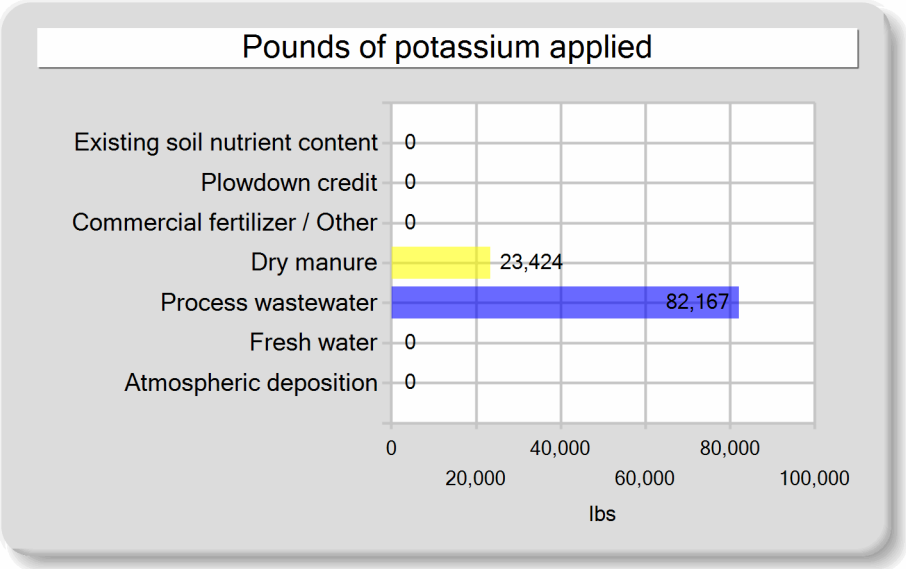
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

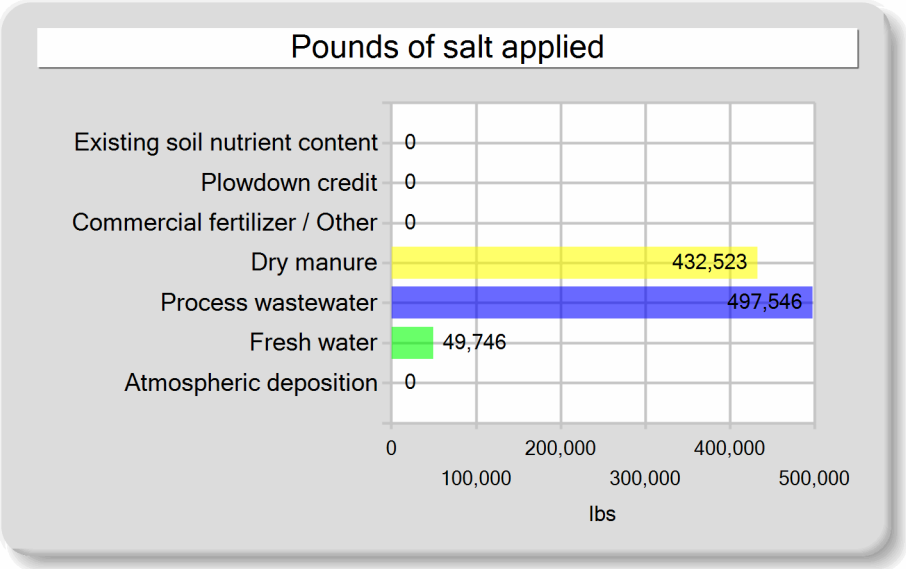


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John De Raadt
John De Raadt (Jun 21, 2024 09:46 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
John and or Joanne DeRaadt	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 21, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest For Existing Milk Cow Dairies

Instructions:

1.) Complete one manifest for **each hauling event,** for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2.) If there are multiple destinations, **complete a separate form for each destination**.
3.) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4.) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: JOHN DERAADT

Name of Dairy Facility: DERAADT DAIRY J.J DERAADT

Facility Address: 6460 SMITH RD (Number and Street) Oakdale (City) 95361 (Zip Code)

Contact Person Name and Phone Number: JOHN DERAADT (Name) 209 840-2572 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: DENIR BROS. AG SERVICES

Address of Hauling Company /Person: 5230 PATTERSON RD (Number and Street) Oakdale (City) 95361 (Zip Code)

Contact Person: ALBERT DENIR (Name) 209 605-1373 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

DENIR BROS AG SERVICES DBA 3D AG (Name)

5230 PATTERSON RD OAKDALE CA 95361 (Number and Street, City, Zip Code, Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

SAME AS ABOVE (Number and Street, City, Zip Code, Assessor's Parcel Number)

Dates Hauled: 5/1 - 5/14/2023

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 1428 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): 1300 # PER YARD

Method used to determine amount of manure:

Truck Load Count

68 LOADS x 21 YARDS PER LOAD = 1428

714 TONS

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: N/A Gallons

Method used to determine volume of process wastewater:

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

_____ Yes X No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

AD (Operator shall provide initials here to acknowledge this requirement).
Initial

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: John De Raadt
John De Raadt (Jun 21, 2024 09:46 PDT) Date: Jun 21, 2024

Hauler's Signature: DR FOR PERIR BROS. Date: 5/14/2023

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
J & J De Raadt Dairy
6460 Smith Rd.
Oakdale, CA 95361

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/11/23 2:27 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 10/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at: 18.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Barn	T3284025A	Electrical Conductivity (EC)	EPA 120.1 and	0.490	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.41	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	313	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)	1.27
Electrical Conductivity (EC)	96.6			0.102
Nitrate Nitrogen (NO3-N)	98.6	97.7 (T3284025A)	96.8 (T3284025A)	0.946

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
J & J De Raadt Dairy
6460 Smith Rd.
Oakdale, CA 95361

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/11/23 2:27 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 10/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at: 19.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Dairy Well	T3284025B	Electrical Conductivity (EC)	EPA 120.1 and	0.173	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.53	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	111	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)	1.27
Electrical Conductivity (EC)	96.6			0.102
Nitrate Nitrogen (NO3-N)	98.6	97.7 (T3284025A)	96.8 (T3284025A)	0.946

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Compliance Analysis Report

Customer
J & J De Raadt Dairy
6460 Smith Rd.
Oakdale, CA 95361

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/11/23 2:27 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 10/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at: 18.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW FLD 2 Angels	T3284025C	Electrical Conductivity (EC)	EPA 120.1 and	0.406	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	7.75	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	260	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)	1.27
Electrical Conductivity (EC)	96.6			0.102
Nitrate Nitrogen (NO3-N)	98.6	97.7 (T3284025A)	96.8 (T3284025A)	0.946

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

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www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
J & J De Raadt Dairy
6460 Smith Rd.
Oakdale, CA 95361

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/11/23 2:27 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 10/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at: 19.0 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW House by Pond	T3284025D	Electrical Conductivity (EC)	EPA 120.1 and	0.410	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.08	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	262	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)	1.27
Electrical Conductivity (EC)	96.6			0.102
Nitrate Nitrogen (NO3-N)	98.6	97.7 (T3284025A)	96.8 (T3284025A)	0.946

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

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Compliance Analysis Report

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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/11/23 2:27 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 10/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at: 19.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW New House	T3284025E	Electrical Conductivity (EC)	EPA 120.1 and	0.365	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.53	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	234	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)	1.27
Electrical Conductivity (EC)	96.6			0.102
Nitrate Nitrogen (NO3-N)	98.6	97.7 (T3284025A)	96.8 (T3284025A)	0.946

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: J & J de Roodt
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F & R Ag
Sampled By: John
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	10/11/23		DW Barn	Domestic	18.2		✓				
2	↓		Dairy New	↓	19.2		↓				
3			FLD 2 Angels		18.4						
4			House by Pond		19.0						
5			New House		19.2						
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	Jennette Savoya	F&R				
2nd		DENELE ANALYTICAL, INC.	10/11/23 14:27			T3284025

Required to be submitted to any agency? (YES / NO)

2023 Annual Report J&J De Raadt

Final Audit Report 2024-06-21

Created:	2024-06-20
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAYN7GqIvNxCJhyiVC6fv9lIFx1GII8KZR

"2023 Annual Report J&J De Raadt" History

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2024-06-20 - 0:13:44 AM GMT

Document emailed to jderaadt@yahoo.com for signature
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Signer jderaadt@yahoo.com entered name at signing as John De Raadt
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